TITLE

Overall and Heterogeneous Effects of Psychotherapy-Informed Push Notifications on Mood, Sleep, and Activity in Medical Interns

AUTHORS

Jack Yang, Andrew Jones, Srijan Sen

CORRESPONDING AUTHOR

1. NAME: Jack Yang

   EMAIL: yhyunjoo@umich.edu

   PHONE: 310-279-0096

INTRODUCTION AND AIMS OF THE STUDY

This section provides an overview of the context of the research question, including what is known and unknown, and requires a detailed literature search. This should contain the following four elements, and will be used to construct the Introduction and Discussion sections of the manuscript:

1. Define the problem or research question.
2. Describe what is known about the research relative to what you plan to study. For example, what studies have already examined an aspect of this research question, and what remaining questions exist that this research project will address.
3. What are the gaps in knowledge that this study will address (usually 2–3).
4. Describe the hypotheses of the study questions.

Depression is among the leading contributors to the global burden of disease [1, 2] and carries a substantial economic cost [3]. Chronic occupational stress is a well-documented contributor to depressive disorders [4], and physicians in training are particularly exposed. Depression and depressive symptoms affect roughly one in four medical students [5, 6], and a pooled 28.8% of resident physicians, with a median absolute increase in depressive symptoms of 15.8% at the onset of residency training [7]. Demanding and unpredictable schedules further limit interns' access to traditional mental health services [8].

Mobile health (mHealth) interventions offer a way to deliver support within daily life. Text-message and handheld-device interventions have produced positive behavior change across a range of health domains

[9, 10, 11, 12, 13], and smartphone-based mental health interventions reduce depressive symptoms [14, 15]. Yet sustained engagement is difficult: treatment burden and fatigue undermine adherence [16], users frequently disengage [17], and the timing and frequency of prompts shape how they are received [18]. Just-in-time adaptive interventions (JITAIs) address these challenges by tailoring the type, timing, and intensity of support to an individual's changing state [19, 20, 21, 22]. Micro-randomized trials (MRTs), which repeatedly randomize intervention delivery within each participant, provide the data needed to determine when and for whom such components are effective [23, 24].

The content of mHealth messages can draw on several established therapeutic strategies, including cognitive therapy [15, 25], behavioral activation [26, 27], mindfulness [28, 29], motivational interviewing [30, 31], and distanced self-talk [32]. The benefits of these strategies are not uniform across individuals. Behavioral activation outperformed cognitive therapy among more severely depressed patients [27]; mindfulness-based cognitive therapy prevented relapse only among patients with three or more prior episodes [28]; and when delivered by smartphone, behavioral activation was more effective for patients with higher initial severity while mindfulness was more effective for those with lower severity [33]. Among medical interns specifically, the effects of push notifications differ across subgroups [34] and depend on recent mood, activity, and sleep [8].

The Intern Health Study (IHS) follows first-year residents throughout internship and embeds MRTs of push notifications targeting mood, physical activity, and sleep. Analyzing a 6-month MRT of 1,565 interns with the weighted and centered least squares (WCLS) estimator [24], NeCamp et al. found that notification effects were moderated by the previous week's mood, step count, and sleep [8]. However, WCLS estimates causal excursion effects averaged across participants and decision points, conditional on a prespecified set of moderators. Heterogeneity that these moderators do not capture is averaged away, so opposing individual responses can cancel and mask meaningful variation in who benefits. Moreover, subsequent IHS cohorts deliver messages based on distinct therapeutic strategies whose individual-level efficacy has not been examined.

The doubly robust WCLS (DR-WCLS) estimator [35] is well suited to this problem. DR-WCLS constructs a doubly robust pseudo-outcome at each decision point, which, because randomization probabilities are known in an MRT, provides an unbiased signal of the treatment effect at that decision. Unlike WCLS, which yields only population-averaged effects, these pseudo-outcomes can be aggregated within each participant to estimate individual-level effects, allowing us to test directly whether responsiveness varies across interns. Leveraging this, we (1) estimate the causal excursion effects of distinct therapeutic notification strategies on proximal mood, sleep, and activity; (2) test whether treatment responsiveness varies significantly across participants; and (3) identify baseline characteristics that predict positive versus negative responsiveness.

We hypothesize that:

H1. Sending a notification improves the proximal outcome in its own domain (mood, sleep, or activity) relative to sending no notification [8].

H2. Effects on the proximal outcome differ by therapeutic strategy.

H3. Treatment responsiveness varies across participants beyond what would be expected by chance under

the randomization design.

H4. Baseline depressive severity, measured by the PHQ-9, moderates the effect of strategy, with behavioral strategy relatively more effective at higher severity and mindfulness relatively more effective at lower severity [27, 33].

H5. Mindfulness messages are more effective for interns with a history of depression [28].

DELIVERABLE TYPE AND DEADLINE (OPTIONAL)

_____	Abstract	Deadline Date: _____
_____	Manuscript	Deadline Date: _____
X	Internal Report	Deadline Date: Oct 21, 2026
_____	Pilot data for grant	Deadline Date: _____
_____	Other (please explain): _____	
	Deadline Date: _____	

STUDY COHORT

PROMPT and COMPASS Cohorts Diagram

☐ RCT PROMPT	☐ R01 PROMPT
☐ COMPASS	☐ Full Cohort
☒ 2025 IHS Cohort	☒ 2024 IHS Cohort

We will prioritize 2024 IHS cohort, possibly extending to the 2025 cohort if time permits, or additional sample size is needed for sufficient power.

DATA TYPES (E.G. SURVEY, FITBIT, GENOMIC, SENSORKIT)

Survey (for BL characteristics), Fitbit & Garmin & HealthKit (for steps/sleep/mood outcomes as well as moderators)

INCLUSIONS (E.G. SCREENING CUTOFFS, APP ASSIGNMENT)

N/A

EXCLUSIONS (IF ANY)

To ensure reliable estimation of participant-level treatment effects, participants with excessive missingness with respect to proximal outcome or moderators will be excluded prior to analysis based on a predefined completion threshold. For retained participants, sporadic decision-point level missingness will be addressed through the observation weighting in the DR-WCLS estimation framework.

VARIABLES

MAIN EXPOSURE

Notification (sending notification vs not sending notification), notification type (sleep vs mood vs activity), notification strategy (therapeutic strategies)

PRIMARY OUTCOME

Self-reported mood, sleep duration, and step count

SECONDARY OUTCOME (S)

N/A

ADDITIONAL COVARIATES (PLEASE SPECIFY IF THESE WILL BE CONSIDERED AS CONFOUNDERS, OR MEDIATING AND MODERATING RELATIONSHIPS WILL ALSO BE EXPLORED)

Additional covariates consist of dynamic, time-varying contextual variables measured at each decision point (e.g., previous night's sleep duration) as well as baseline characteristics. These variables will be evaluated as candidate moderators to assess contextual heterogeneity in treatment responsiveness and determine how real-time user state and characteristics affect the intervention effect on the proximal outcome.

STATISTICAL ANALYSIS

This can be your fully established analysis, or draft prior to potential consult with our analysis team.

Univariate approach:

Multivariable approach:

We evaluate overall treatment efficacy and individual response heterogeneity using a multi-stage causal inference framework. We begin by applying Doubly Robust Weighted and Centered Least Squares to estimate the overall marginal causal excursion effect across all participants and decision points. Next, using the per-decision-point doubly robust pseudo-outcomes generated by this procedure, we fit participant-specific regressions using time-varying contextual variables to estimate individual baseline treatment effects. To establish whether these individual treatment effects are statistically significant overall across the population, we compare observed effect magnitudes against a randomization null distribution generated by permuting treatment assignments under the design probabilities. We then inspect the distribution of estimated participant effects and fit Gaussian Mixture Models to evaluate response heterogeneity and identify distinct responder subgroups. Finally, if significant heterogeneity is confirmed, we regress participant baseline characteristics against individual treatment effects to determine which baseline profiles predict positive versus negative intervention responsiveness.

All analyses will be conducted across the complete study sample as well as stratified by participant device type to evaluate potential hardware-level variability in treatment estimates.

LIMITATIONS OF THE STUDY (OPTIONAL)

A primary limitation of this study concerns the preprocessing and harmonization of intensive time-series data across heterogeneous commercial devices. Because participants were not explicitly differentiated by device model, establishing uniform measurement baselines across hardware platforms proved challenging. Consequently, cross-device data harmonization relied on ad-hoc alignment procedures, which may introduce calibration noise or unmeasured sensor variance. On the other hand, device stratification will reduce the power of the analysis. Furthermore, the sensitivity of causal estimates to specific time-series preprocessing choices remains a potential source of analytical variability.

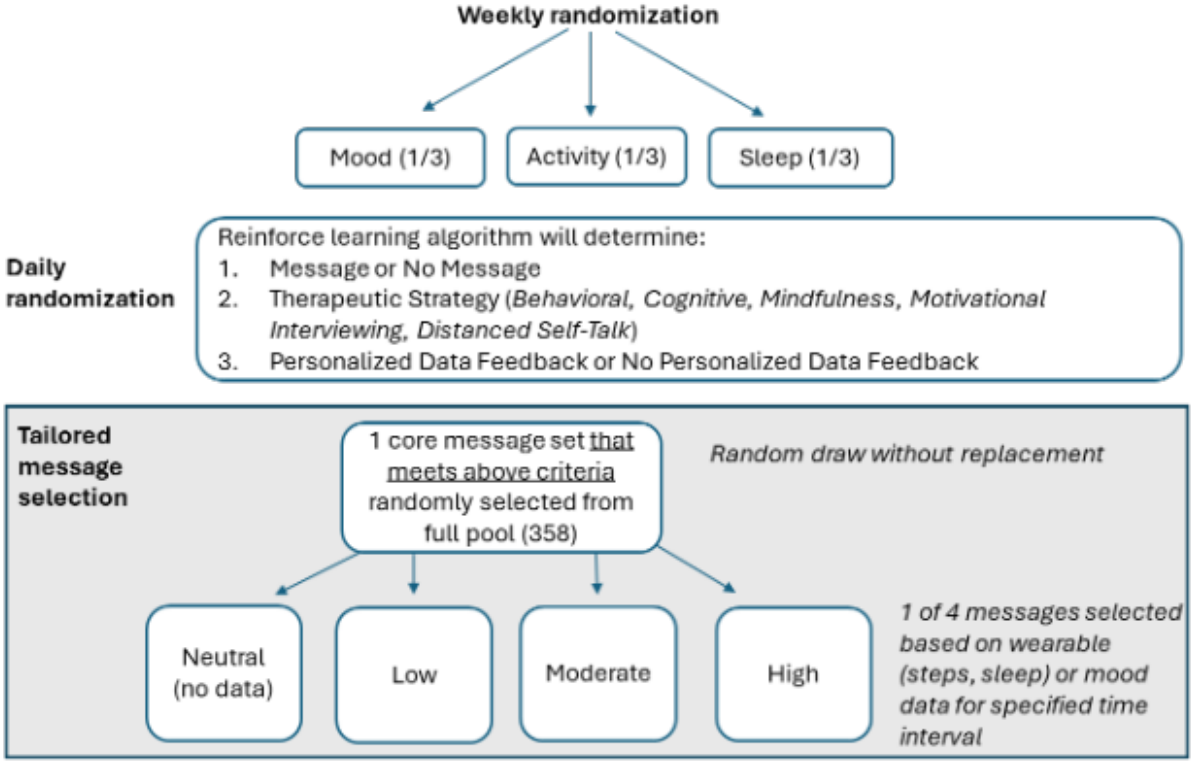
TABLES/ FIGURES

Title, description and shell of tables and mock figures. The following are examples for Table 1 and figures.

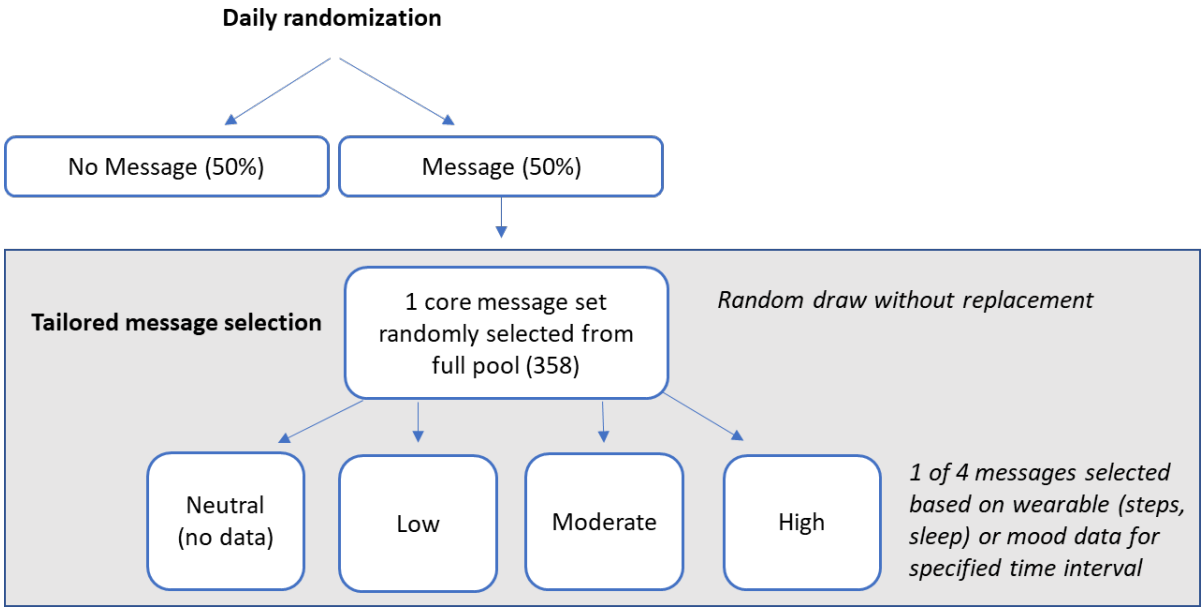
1. Table 1 – Characteristics

Characteristic	Overall (No. of Patients, %)	Garmin (No. Patients, % Total)	HealthKit (No. Patients, % Total)	Fitbit (No. Patients, % Total)
Race				
White				
Black				
Others				
Age				
…	…	…	…	…

2. Figure 1 – 2025 IHS Study Design



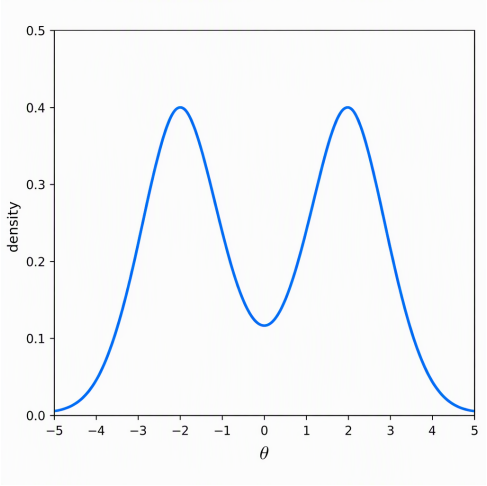
3. Figure 2 – 2024 IHS Study Design



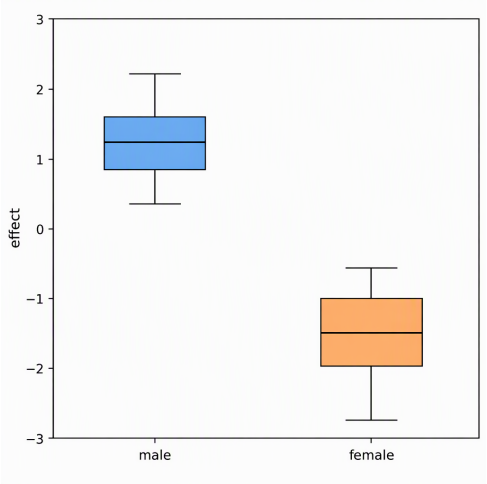
4. Table 2 – Table of Cumulative Magnitude of Effects and P-Values

Intervention	Cumulative Magnitude of Effect	P-Value
Notification		
Mindfulness		
…	…	…
No Notification		

5. Figure 3 – Distribution of Effects



6. Figure 4 – Effect vs Baseline Characteristic



REFERENCES

[1] Harvey Whiteford et al. “Global burden of disease attributable to mental and substance use disorders: findings from the Global Burden of Disease Study 2010”. In: *The Lancet* (2013). DOI: 10.1016/s0140-6736(13)61611-6. URL: http://doi.org/10.1016/s0140-6736(13)61611-6.

[2] GBD 2019 Mental Disorders Collaborators. “Global, regional, and national burden of 12 mental disorders in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019”. en. In: *The Lancet Psychiatry* 9.2 (Feb. 2022), pp. 137–150. ISSN: 22150366. DOI: 10.1016/S2215-0366(21)00395-3. URL: https://linkinghub.elsevier.com/retrieve/pii/S2215036621003953 (visited on 10/14/2026).

[3] Paul E. Greenberg et al. “The Economic Burden of Adults With Major Depressive Disorder in the United States (2005 and 2010)”. In: *The Journal of Clinical Psychiatry* (2015). DOI: 10.4088/jcp.14m09298. URL: http://doi.org/10.4088/jcp.14m09298.

[4] Christopher Tennant. “Work-related stress and depressive disorders”. In: *Journal of Psychosomatic Research* 51.5 (Nov. 2001), pp. 697–704. ISSN: 0022-3999. DOI: 10.1016/S0022-3999(01)00255-0. URL: https://www.sciencedirect.com/science/article/pii/S0022399901002550 (visited on 10/13/2026).

[5] Lisa S. Rotenstein et al. “Prevalence of Depression, Depressive Symptoms, and Suicidal Ideation Among Medical Students: A Systematic Review and Meta-Analysis”. en. In: *JAMA* 316.21 (Dec. 2016), p. 2214. ISSN: 0098-7484. DOI: 10.1001/jama.2016.17324. URL: http://jama.jamanetwork.com/article.aspx?doi=10.1001/jama.2016.17324 (visited on 10/08/2026).

[6] Liselotte N. Dyrbye, Matthew R. Thomas, and Tait D. Shanafelt. “Systematic Review of Depression, Anxiety, and Other Indicators of Psychological Distress Among U.S. and Canadian Medical Students”. In: *Academic Medicine* (2006). DOI: 10.1097/00001888-200604000-00009. URL: http://doi.org/10.1097/00001888-200604000-00009.

[7] Douglas A. Mata et al. “Prevalence of Depression and Depressive Symptoms Among Resident Physicians A Systematic Review and Meta-analysis”. In: *JAMA* 314.22 (Dec. 2015), pp. 2373–2383. ISSN: 0098-7484. DOI: 10.1001/jama.2015.15845. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC4866499/ (visited on 10/15/2026).

[8] Timothy NeCamp et al. “Assessing Real-Time Moderation for Developing Adaptive Mobile Health Interventions for Medical Interns: Micro-Randomized Trial”. In: *Journal of Medical Internet Research* 22.3 (Mar. 2020), e15033. ISSN: 1439-4456. DOI: 10.2196/15033. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC7157494/ (visited on 10/15/2026).

[9] Brianna S. Fjeldsoe, Alison L. Marshall, and Yvette D. Miller. “Behavior Change Interventions Delivered by Mobile Telephone Short-Message Service”. In: *American Journal of Preventive Medicine* 36.2 (Feb. 2009), pp. 165–173. ISSN: 0749-3797. DOI: 10.1016/j.amepre.2008.09.040. URL: https://www.sciencedirect.com/science/article/pii/S0749379708008921 (visited on 10/14/2026).

[10] Elaine De Leon, Laura W Fuentes, and Joanna E Cohen. “Characterizing Periodic Messaging Interventions Across Health Behaviors and Media: Systematic Review”. en. In: *Journal of Medical Internet Research* 16.3 (Mar. 2014), e93. ISSN: 14388871. DOI: 10.2196/jmir.2837. URL: http://www.jmir.org/2014/3/e93/ (visited on 10/14/2026).

[11] Kevin Patrick et al. “A Text Message–Based Intervention for Weight Loss: Randomized Controlled Trial”. en. In: *Journal of Medical Internet Research* 11.1 (Jan. 2009), e1. ISSN: 1438-8871. DOI: 10.2196/jmir.1100. URL: http://www.jmir.org/2009/1/e1/ (visited on 10/14/2026).

[12] Victoria L. Franklin et al. “A randomized controlled trial of Sweet Talk, a text-messaging system to support young people with diabetes”. In: *Diabetic Medicine* (2006). DOI: 10.1111/j.1464-5491.2006.01989.x. URL: http://doi.org/10.1111/j.1464-5491.2006.01989.x.

[13] Abby C. King et al. “Promoting Physical Activity Through Hand-Held Computer Technology”. In: *American Journal of Preventive Medicine* 34.2 (Feb. 2008), pp. 138–142. ISSN: 0749-3797. DOI: 10.1016/j.amepre.2007.09.025. URL: https://www.sciencedirect.com/science/article/pii/S0749379707006459 (visited on 10/14/2026).

[14] Joseph Firth et al. “The efficacy of smartphone-based mental health interventions for depressive symptoms: a meta-analysis of randomized controlled trials”. en. In: *World Psychiatry* 16.3 (2017), pp. 287–298. ISSN: 2051-5545. DOI: 10.1002/wps.20472. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/wps.20472 (visited on 10/15/2026).

[15] Jake Linardon et al. “The efficacy of app-supported smartphone interventions for mental health problems: a meta-analysis of randomized controlled trials”. en. In: *World Psychiatry* 18.3 (2019), pp. 325–336. ISSN: 2051-5545. DOI: 10.1002/wps.20673. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/wps.20673 (visited on 10/15/2026).

[16] Bryan W Heckman, Amanda R Mathew, and Matthew J Carpenter. “Treatment burden and treatment fatigue as barriers to health”. In: *Current Opinion in Psychology*. Health behavior 5 (Oct. 2015), pp. 31–36. ISSN: 2352-250X. DOI: 10.1016/j.copsyc.2015.03.004. URL: https://www.sciencedirect.com/science/article/pii/S2352250X15001098 (visited on 10/14/2026).

[17] Shaodian Zhang and Noémie Elhadad. “Factors Contributing to Dropping-out in an Online Health Community: Static and Longitudinal Analyses”. In: *AMIA Annual Symposium Proceedings* 2016 (Feb. 2017), pp. 2090–2099. ISSN: 1942-597X. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC5333218/ (visited on 10/14/2026).

[18] Leanne G. Morrison et al. “The Effect of Timing and Frequency of Push Notifications on Usage of a Smartphone-Based Stress Management Intervention: An Exploratory Trial”. en. In: *PLOS ONE* 12.1 (Jan. 2017), e0169162. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0169162. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0169162 (visited on 10/08/2026).

[19] Inbal Billie Nahum-Shani et al. “Just-in-Time Adaptive Interventions (JITAIs) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support”. In: *Annals of Behavioral Medicine* (2016). DOI: 10.1007/s12160-016-9830-8. URL: http://doi.org/10.1007/s12160-016-9830-8.

[20] Kristin E. Heron and Joshua M. Smyth. “Ecological momentary interventions: Incorporating mobile technology into psychosocial and health behaviour treatments”. en. In: *British Journal of Health Psychology* 15.1 (Feb. 2010), pp. 1–39. ISSN: 2044-8287. DOI: 10.1348/135910709X466063. URL: https://bpspsychub.onlinelibrary.wiley.com/doi/10.1348/135910709X466063 (visited on 10/14/2026).

[21] William T. Riley et al. “Health behavior models in the age of mobile interventions: are our theories up to the task?” In: *Translational Behavioral Medicine* (2011). DOI: 10.1007/s13142-011-0021-7. URL: http://doi.org/10.1007/s13142-011-0021-7.

[22] Liyuan Wang and Lynn Carol Miller. “Just-in-the-Moment Adaptive Interventions (JITAI): A Meta-Analytical Review”. In: *Health Communication* (2019). DOI: 10.1080/10410236.2019.1652388. URL: http://doi.org/10.1080/10410236.2019.1652388.

[23] Tianchen Qian et al. “The Micro-Randomized Trial for Developing Digital Interventions: Experimental Design and Data Analysis Considerations”. In: *arXiv (Cornell University)* (2021). DOI: 10.48550/arxiv.2107.03544. URL: http://doi.org/10.48550/arxiv.2107.03544.

[24] Audrey Boruvka et al. “Assessing Time-Varying Causal Effect Moderation in Mobile Health”. In: *Journal of the American Statistical Association* (2017). DOI: 10.1080/01621459.2017.1305274. URL: http://doi.org/10.1080/01621459.2017.1305274.

[25] Aaron T. Beck et al. *Cognitive Therapy of Depression*. 2nd ed. New York, UNITED STATES: Guilford Publications, 2024. ISBN: 978-1-4625-5450-8. URL: http://ebookcentral.proquest.com/lib/umichigan/detail.action?docID=31319082 (visited on 10/15/2026).

[26] Peter M. Lewinsohn. “A behavioral approach to depression”. In: *The psychology of depression: Contemporary theory and research*. Oxford, England: John Wiley & Sons, 1974. ISBN: 978-0-470-28084-3.

[27] Sona Dimidjian et al. “Randomized trial of behavioral activation, cognitive therapy, and antidepressant medication in the acute treatment of adults with major depression.” eng. In: *Journal of Consulting and Clinical Psychology* 74.4 (Aug. 2006), pp. 658–670. ISSN: 1939-2117. DOI: 10.1037/0022-006X.74.4.658. URL: https://research.ebsco.com/plink/24702d3b-1b8d-3a7d-93fd-0911d11a5f82 (visited on 10/15/2026).

[28] John D. Teasdale et al. “Prevention of relapse/recurrence in major depression by mindfulness-based cognitive therapy.” eng. In: *Journal of Consulting and Clinical Psychology* 68.4 (Aug. 2000), pp. 615–623. ISSN: 1939-2117. DOI: 10.1037/0022-006X.68.4.615. URL: https://research.ebsco.com/plink/6727d7f9-8e6f-3936-92e7-49493016d703 (visited on 10/15/2026).

[29] Marcos Economides et al. “Improvements in Stress, Affect, and Irritability Following Brief Use of a Mindfulness-based Smartphone App: A Randomized Controlled Trial”. In: *Mindfulness* 9.5 (2018), pp. 1584–1593. ISSN: 1868-8527. DOI: 10.1007/s12671-018-0905-4. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC6153897/ (visited on 10/15/2026).

[30] Sune Rubak et al. “Motivational interviewing: a systematic review and meta-analysis”. In: *The British Journal of General Practice* 55.513 (Apr. 2005), pp. 305–312. ISSN: 0960-1643. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC1463134/ (visited on 10/15/2026).

[31] Rebecca M. Shingleton and Tibor P. Palfai. “Technology-delivered adaptations of motivational interviewing for health-related behaviors: A systematic review of the current research”. en. In: *Patient Education and Counseling* 99.1 (Jan. 2016), pp. 17–35. ISSN: 07383991. DOI: 10.1016/j.pec.2015.08.005. URL: https://linkinghub.elsevier.com/retrieve/pii/S0738399115300434 (visited on 10/15/2026).

[32] E. Kross and O. Ayduk. “Self-Distancing”. en. In: *Advances in Experimental Social Psychology*. Vol. 55. Elsevier, 2017, pp. 81–136. ISBN: 978-0-12-812115-3. DOI: 10.1016/bs.aesp.2016.10.002. URL: https://linkinghub.elsevier.com/retrieve/pii/S0065260116300338 (visited on 10/15/2026).

[33] Kien Hoa Ly et al. “Behavioural activation versus mindfulness-based guided self-help treatment administered through a smartphone application: a randomised controlled trial”. en. In: *BMJ Open* 4.1 (Jan. 2014), e003440. ISSN: 2044-6055, 2044-6055. DOI: 10.1136/bmjopen-2013-003440. URL: https://bmjopen.bmj.com/lookup/doi/10.1136/bmjopen-2013-003440 (visited on 10/15/2026).

[34] Jiaai Xu et al. “Uncovering the Varied Impact of Behavioral Change Messages on Population Groups”. en. In: *Proceedings of the 8th Machine Learning for Healthcare Conference*. PMLR, Dec. 2023, pp. 906–922. URL: https://proceedings.mlr.press/v219/xu23a.html (visited on 10/02/2026).

[35] Jieru Shi and Walter Dempsey. “A meta-learning method for estimation of causal excursion effects to assess time-varying moderation”. In: *Biometrics* 81.4 (Dec. 2025), ujaf129. ISSN: 1541-0420. DOI: 10.1093/biomtc/ujaf129. URL: https://doi.org/10.1093/biomtc/ujaf129 (visited on 10/01/2026).